

Managing Deployments Using Kubernetes Engine - LAB

GSP053



Overview

Dev Ops practices will regularly make use of multiple deployments to manage application deployment scenarios such as "Continuous deployment", "Blue-Green deployments", "Canary deployments" and more. This lab teaches you how to scale and manage containers so you can accomplish these common scenarios where multiple heterogeneous deployments are being used.

Objectives

In this lab, you will learn how to perform the following tasks:

- Use the kubectl tool
- Create deployment yaml files
- Launch, update, and scale deployments
- Update deployments and learn about deployment styles



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End Lab00:57:07

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud Console

Username

student-02-e14d37b92bd3f📋

Password

Rf6aKqhuIWHB📋

Project ID

qwiklabs-gcp-04-f2da343f📋

3. (Optional) You can list the active account name with this command:

gcloud auth list📋

4. Click **Authorize**.

Output:

ACTIVE: *
ACCOUNT: student-02-e14d37b92bd3@qwiklabs.net

To set the active account, run:
\$ gcloud config set account 'ACCOUNT'

5. (Optional) You can list the project ID with this command:

gcloud config list project📋

Output:

0/100

deployments

Setup and requirements

Set the zone

Get sample code for this lab

Task 1. Learn about the deployment object

Task 2. Create a deployment

Task 3. Rolling update

Task 4. Canary deployments

Task 5. Blue-green deployments

Congratulations!

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Open Google Cloud Console

Username

student-02-e14d37b92bd3i

Password

Rf6aKqhuIWHB

Project ID

qwiklabs-gcp-04-f2da343i

Set the zone

Set your working Google Cloud zone by running the following command, substituting the local zone as `us-west1-b`:

```
gcloud config set compute/zone us-west1-b
```

Get sample code for this lab

1. Get the sample code for creating and running containers and deployments from the lab bucket:

```
gcloud storage cp -r gs://spl/gsp053/kubernetes .
cd kubernetes
```

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Username

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Password

Rf6aKqhuIWHB

Project ID

qwiklabs-gcp-04-f2da343i

Get sample code for this lab

1. Get the sample code for creating and running containers and deployments from the lab bucket:

```
gcloud storage cp -r gs://spl/s/gsp053/kubernetes .
cd kubernetes
```

2. Create a cluster with 3 nodes (this will take a few minutes to complete):

```
gcloud container clusters create bootcamp \
  --machine-type e2-small \
  --num-nodes 3 \
  --scopes
"https://www.googleapis.com/auth/projecthosting,storage-rw"
```

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Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to **qwiklabs-gcp-04-f2da343063b3**.
Use ``gcloud config set project [PROJECT_ID]`` to change to a different project.
student_02_e14d37b92bd3@cloudshell:~ (**qwiklabs-gcp-04-f2da343063b3**)\$ `gcloud auth list`
Credentialed Accounts

ACTIVE: *
ACCOUNT: student-02-e14d37b92bd3@qwiklabs.net

To set the active account, run:
\$ `gcloud config set account 'ACCOUNT'`

student_02_e14d37b92bd3@cloudshell:~ (**qwiklabs-gcp-04-f2da343063b3**)\$ `gcloud config list project`
INFORMATION: Project 'qwiklabs-gcp-04-f2da343063b3' has no 'environment' tag set. Use either 'Production', 'Development', 'Test', or 'Staging'. Add an 'environment' tag using ``gcloud resource -manager tags bindings create``.
[core]
project = qwiklabs-gcp-04-f2da343063b3

Your active configuration is: [cloudshell-5202]
student_02_e14d37b92bd3@cloudshell:~ (**qwiklabs-gcp-04-f2da343063b3**)\$ `gcloud config set compute/zone us-west1-b`
Updated property [compute/zone].
student_02_e14d37b92bd3@cloudshell:~ (**qwiklabs-gcp-04-f2da343063b3**)\$ `gcloud config set compute/zone us-west1-b`
Updated property [compute/zone].
student_02_e14d37b92bd3@cloudshell:~ (**qwiklabs-gcp-04-f2da343063b3**)\$ `gcloud storage cp -r gs://spl/gsp053/kubernetes .`
`cd kubernetes`
Copying `gs://spl/gsp053/kubernetes/cleanup.sh` to `file:///kubernetes/cleanup.sh`
Copying `gs://spl/gsp053/kubernetes/deployments/fortune-app-blue.yaml` to `file:///kubernetes/deployments/fortune-app-blue.yaml`
Copying `gs://spl/gsp053/kubernetes/deployments/fortune-app-canary.yaml` to `file:///kubernetes/deployments/fortune-app-canary.yaml`
Copying `gs://spl/gsp053/kubernetes/deployments/fortune-app-green.yaml` to `file:///kubernetes/deployments/fortune-app-green.yaml`
Copying `gs://spl/gsp053/kubernetes/services/fortune-app-blue-service.yaml` to `file:///kubernetes/services/fortune-app-blue-service.yaml`
Copying `gs://spl/gsp053/kubernetes/services/fortune-app-green-service.yaml` to `file:///kubernetes/services/fortune-app-green-service.yaml`
Copying `gs://spl/gsp053/kubernetes/services/fortune-app.yaml` to `file:///kubernetes/services/fortune-app.yaml`

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Copying gs://spl/spl/gsp053/kubernetes/deployments/fortune-app-canary.yaml to file:///kubernetes/deployments/fortune-app-canary.yaml

Copying gs://spl/spl/gsp053/kubernetes/deployments/fortune-app-green.yaml to file:///kubernetes/deployments/fortune-app-green.yaml

Copying gs://spl/spl/gsp053/kubernetes/services/fortune-app-blue-service.yaml to file:///kubernetes/services/fortune-app-blue-service.yaml

Copying gs://spl/spl/gsp053/kubernetes/services/fortune-app-green-service.yaml to file:///kubernetes/services/fortune-app-green-service.yaml

Copying gs://spl/spl/gsp053/kubernetes/services/fortune-app.yaml to file:///kubernetes/services/fortune-app.yaml

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Copying gs://spl/spl/gsp053/kubernetes/tls/ca.pem to file:///kubernetes/tls/ca.pem

Copying gs://spl/spl/gsp053/kubernetes/tls/cert.pem to file:///kubernetes/tls/cert.pem

Copying gs://spl/spl/gsp053/kubernetes/tls/key.pem to file:///kubernetes/tls/key.pem

Completed files 11/11 | 9.5kiB/9.5kiB

Average throughput: 11.8kiB/s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)\$ gcloud container clusters create bootcamp \

--machine-type e2-small \

--num-nodes 3 \

--scopes "https://www.googleapis.com/auth/projecthosting,storage-rw"

Note: Your Pod address range (--cluster-ipv4-cidr) can accommodate at most 1008 node(s).

Creating cluster bootcamp in us-west1-b... Cluster is being health-checked (Kubernetes Control Plane is healthy)...done.

Created [https://container.googleapis.com/v1/projects/qwiklabs-gcp-04-f2da343063b3/zones/us-west1-b/clusters/bootcamp].

To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/workload/_gcloud/us-west1-b/bootcamp?project=qwiklabs-gcp-04-f2da343063b3

kubeconfig entry generated for bootcamp.

NAME: bootcamp

LOCATION: us-west1-b

MASTER_VERSION: 1.33.5-gke.1201000

MASTER_IP: 34.11.173.99

MACHINE_TYPE: e2-small

NODE_VERSION: 1.33.5-gke.1201000

NUM_NODES: 3

STATUS: RUNNING

STACK_TYPE: IPV4

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)\$

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Task 1. Learn about the deployment object

To get started, take a look at the deployment object.

1. The `explain` command in `kubectl` can tell us about the deployment object:

```
kubectl explain deployment
```

2. You can also see all of the fields using the `--recursive` option:

```
kubectl explain deployment --recursive
```

3. You can use the `explain` command as you go through the lab to help you understand the structure of a deployment object and understand what the individual fields do:

```
kubectl explain deployment.metadata.name
```

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```
NUM_NODES: 3
STATUS: RUNNING
STACK_TYPE: IPV4
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl explain deployment
GROUP:      apps
KIND:       Deployment
VERSION:    v1

DESCRIPTION:
  Deployment enables declarative updates for Pods and ReplicaSets.

FIELDS:
  apiVersion    <string>
    APIVersion defines the versioned schema of this representation of an object.
    Servers should convert recognized schemas to the latest internal value, and
    may reject unrecognized values. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources

  kind    <string>
    Kind is a string value representing the REST resource this object
    represents. Servers may infer this from the endpoint the client submits
    requests to. Cannot be updated. In CamelCase. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds

  metadata    <ObjectMeta>
    Standard object's metadata. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata

  spec    <DeploymentSpec>
    Specification of the desired behavior of the Deployment.

  status    <DeploymentStatus>
    Most recently observed status of the Deployment.

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$
```

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```
      values      <[]string>
      matchLabels <map[string]string>
      namespaces  <[]string>
      topologyKey <string> -required-
      automountServiceAccountToken <boolean>
      containers  <[]Container> -required-
      args <[]string>
      command     <[]string>
      env <[]EnvVar>
        name      <string> -required-
        value     <string>
        valueFrom <EnvVarSource>
          configMapKeyRef <ConfigMapKeySelector>
            key <string> -required-
            name <string>
            optional <boolean>
          fieldRef <ObjectFieldSelector>
            apiVersion <string>
            fieldPath <string> -required-
          resourceFieldRef <ResourceFieldSelector>
            containerName <string>
            divisor <Quantity>
            resource <string> -required-
          secretKeyRef <SecretKeySelector>
            key <string> -required-
            name <string>
            optional <boolean>
        envFrom <[]EnvFromSource>
          configMapRef <ConfigMapEnvSource>
            name <string>
            optional <boolean>
          prefix <string>
          secretRef <SecretEnvSource>
            name <string>
            optional <boolean>
      image <string>
```

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```
storagePolicyName <string>
volumePath <string> -required-
status <DeploymentStatus>
availableReplicas <integer>
collisionCount <integer>
conditions <[]DeploymentCondition>
  lastTransitionTime <string>
  lastUpdateTime <string>
  message <string>
  reason <string>
  status <string> -required-
  type <string> -required-
observedGeneration <integer>
readyReplicas <integer>
replicas <integer>
terminatingReplicas <integer>
unavailableReplicas <integer>
updatedReplicas <integer>
```

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl explain deployment.metadata.name

GROUP: apps
KIND: Deployment
VERSION: v1

FIELD: name <string>

DESCRIPTION:
Name must be unique within a namespace. Is required when creating resources, although some resources may allow a client to request the generation of an appropriate name automatically. Name is primarily intended for creation idempotence and configuration definition. Cannot be updated. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names#names>

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$

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Task 2. Create a deployment

1. Create the `fortune-app` deployment. Examine the deployment configuration file:

```
cat deployments/fortune-app-blue.yaml
```

Output:

```
# orchestrate-with-kubernetes/kubernetes/deployments/fortune-app-
blue.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: fortune-app-blue
spec:
  replicas: 3
  selector:
    matchLabels:
      app: fortune-app
  template:
```

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https://kubernetes.io/docs/concepts/overview/working-with-objects/names#names

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)\$ cat deployments/fortune-app-blue.yaml

orchestrate-with-kubernetes/kubernetes/deployments/fortune-app-blue.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

name: fortune-app-blue

spec:

replicas: 3

selector:

matchLabels:

app: fortune-app

template:

metadata:

labels:

app: fortune-app

track: stable

version: "1.0.0"

spec:

containers:

- name: fortune-app

The new, centralized image path

image: "us-central1-docker.pkg.dev/qwiklabs-resources/spl-lab-apps/fortune-service:1.0.0"

ports:

- name: http

containerPort: 8080

env:

- name: APP_VERSION

value: "1.0.0"

resources:

limits:

cpu: "0.2"

memory: "20Mi"

readinessProbe:

httpGet:

path: /

port: 8080

initialDelaySeconds: 5

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)\$

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```
containers:
- name: fortune-app
  # The new, centralized image path
  image: "us-central1-docker.pkg.dev/qwiklabs-resources/spl-lab-apps/fortune-service:1.0.0"
  ports:
  - name: http
    containerPort: 8080
  env:
  - name: APP_VERSION
    value: "1.0.0"
  resources:
    limits:
      cpu: "0.2"
      memory: "20Mi"
  readinessProbe:
    httpGet:
      path: /
      port: 8080
    initialDelaySeconds: 5

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl create -f deployments/fortune-app-blue.yaml
deployment.apps/fortune-app-blue created
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
fortune-app-blue 0/3      3            0           7s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
fortune-app-blue 0/3      3            0          10s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get replicasets
NAME          DESIRED   CURRENT   READY   AGE
fortune-app-blue-5cd9c5fd96 3          3          3       23s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
fortune-app-blue-5cd9c5fd96-2pr76 1/1     Running   0          32s
fortune-app-blue-5cd9c5fd96-ng57r 1/1     Running   0          32s
fortune-app-blue-5cd9c5fd96-x9dws 1/1     Running   0          32s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
```

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Username

student-02-e14d37b92bd36

Password

Rf6aKqhuIWHB

Project ID

```
qwiklabs-gcp-04-f2da343d
```

6. Now, create a service to expose the `fortune-app` deployment externally.

```
kubectl create -f services/fortune-app.yaml
```

7. Interact with the `fortune-app` by grabbing its external IP and then curling the `/version` endpoint:

```
kubectl get services fortune-app
```

Note: It may take a few seconds before the External-IP field is populated for your service. This is normal. Just re-run the above command every few seconds until the field is populated.

```
curl http://<EXTERNAL-IP>/version
```

You should get back a JSON response indicating `{"version": "1.0.0"}`.

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Congratulations!

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```
readinessProbe:
  httpGet:
    path: /
    port: 8080
  initialDelaySeconds: 5
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl create -f deployments/fortune-app-blue.yaml
deployment.apps/fortune-app-blue created
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
fortune-app-blue 0/3      3            0           7s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
fortune-app-blue 0/3      3            0           10s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get replicasets
NAME          DESIRED   CURRENT   READY   AGE
fortune-app-blue-5cd9c5fd96 3          3          3       23s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
fortune-app-blue-5cd9c5fd96-2pr76 1/1     Running   0           32s
fortune-app-blue-5cd9c5fd96-ng57r 1/1     Running   0           32s
fortune-app-blue-5cd9c5fd96-x9dws 1/1     Running   0           32s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl create -f services/fortune-app.yaml
service/fortune-app created
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27 <pending>     80:31269/TCP     10s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27 <pending>     80:31269/TCP     14s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27 <pending>     80:31269/TCP     16s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27 34.11.254.223 80:31269/TCP     39s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
```

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Open Google Cloud Console

Username

student-02-e14d37b92bd3i

Password

Rf6aKqhuIWHB

Project ID

qwiklabs-gcp-04-f2da343i

kubectl get services fortune-app

Note: It may take a few seconds before the External-IP field is populated for your service. This is normal. Just re-run the above command every few seconds until the field is populated.

curl http://<EXTERNAL-IP>/version

You should get back a JSON response indicating {"version": "1.0.0"}.

8. You can also use the output templating feature of kubectl to use curl as a one-liner:

curl http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version

Test completed task

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Congratulations!

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```
port: 8080
initialDelaySeconds: 5
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl create -f deployments/fortune-app-blue.yaml
deployment.apps/fortune-app-blue created
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
fortune-app-blue 0/3      3             0           7s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
fortune-app-blue 0/3      3             0           10s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl get replicasets
NAME          DESIRED   CURRENT   READY   AGE
fortune-app-blue-5cd9c5fd96 3          3          3       23s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
fortune-app-blue-5cd9c5fd96-2pr76 1/1     Running   0           32s
fortune-app-blue-5cd9c5fd96-ng57r 1/1     Running   0           32s
fortune-app-blue-5cd9c5fd96-x9dws 1/1     Running   0           32s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl create -f services/fortune-app.yaml
service/fortune-app created
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27    <pending>         80:31269/TCP     10s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27    <pending>         80:31269/TCP     14s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27    <pending>         80:31269/TCP     16s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27    34.11.254.223    80:31269/TCP     39s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ curl http://<EXTERNAL-IP>/^CrSION
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ curl http://34.11.254.223/version
{"version":"1.0.0"}
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$
```

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student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl create -f deployments/fortune-app-blue.yaml

deployment.apps/fortune-app-blue created

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get deployments

NAME READY UP-TO-DATE AVAILABLE AGE

fortune-app-blue 0/3 3 0 7s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get deployments

NAME READY UP-TO-DATE AVAILABLE AGE

fortune-app-blue 0/3 3 0 10s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get replicasets

NAME DESIRED CURRENT READY AGE

fortune-app-blue-5cd9c5fd96 3 3 3 23s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get pods

NAME READY STATUS RESTARTS AGE

fortune-app-blue-5cd9c5fd96-2pr76 1/1 Running 0 32s

fortune-app-blue-5cd9c5fd96-ng57r 1/1 Running 0 32s

fortune-app-blue-5cd9c5fd96-x9dws 1/1 Running 0 32s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl create -f services/fortune-app.yaml

service/fortune-app created

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get services fortune-app

NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE

fortune-app LoadBalancer 34.118.233.27 <pending> 80:31269/TCP 10s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get services fortune-app

NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE

fortune-app LoadBalancer 34.118.233.27 <pending> 80:31269/TCP 14s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get services fortune-app

NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE

fortune-app LoadBalancer 34.118.233.27 <pending> 80:31269/TCP 16s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get services fortune-app

NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE

fortune-app LoadBalancer 34.118.233.27 34.11.254.223 80:31269/TCP 39s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://<EXTERNAL-IP>/^CrSION

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://34.11.254.223/version

{"version":"1.0.0"}

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version

{"version":"1.0.0"}

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$

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End Lab

00:42:44

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[Learn more.](#)

[Open Google Cloud Console](#)

Username

student-02-e14d37b92bd30

Password

Rf6aKqhuIWHB

Project ID

gwiklabs-qcp-04-f2da3430

Scale a deployment

Now that you have a deployment created, you can scale it. Do this by updating the `spec.replicas` field.

1. The replicas field can be most easily updated using the `kubectl scale` command:

```
kubectl scale deployment fortune-app-blue --replicas=5
```

Note: It may take a minute or so for all the new pods to start up.

2. Verify that there are now 5 `fortune-app-blue` Pods running:

```
kubectl get pods | grep fortune-app-blue | wc -l
```

- ### 3. Now scale back the application:

```
kubectl scale deployment fortune-app-blue --replicas=3
```

deployments

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Get sample code for this lab

Task 1. Learn about the deployment object

Task 2. Create a deployment

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Task 4. Canary deployments

Task 5. Blue-green deployments

Congratulations!

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```
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get replicasets
NAME                               DESIRED   CURRENT   READY   AGE
fortune-app-blue-5cd9c5fd96        3         3         3       23s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods
NAME                               READY     STATUS    RESTARTS   AGE
fortune-app-blue-5cd9c5fd96-2pr76  1/1       Running   0          32s
fortune-app-blue-5cd9c5fd96-ng57r  1/1       Running   0          32s
fortune-app-blue-5cd9c5fd96-x9dws  1/1       Running   0          32s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl create -f services/fortune-app.yaml
service/fortune-app created
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
fortune-app  LoadBalancer  34.118.233.27 <pending>     80:31269/TCP    10s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
fortune-app  LoadBalancer  34.118.233.27 <pending>     80:31269/TCP    14s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
fortune-app  LoadBalancer  34.118.233.27 <pending>     80:31269/TCP    16s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
fortune-app  LoadBalancer  34.118.233.27 34.11.254.223 80:31269/TCP    39s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ curl http://<EXTERNAL-IP>/^CrSION
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ curl http://34.11.254.223/version
{"version":"1.0.0"}
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ curl http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version
{"version":"1.0.0"}
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl scale deployment fortune-app-blue --replicas=5
deployment.apps/fortune-app-blue scaled
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods | grep fortune-app-blue | wc -l
5
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl scale deployment fortune-app-blue --replicas=3
deployment.apps/fortune-app-blue scaled
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
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https://console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-f2da343063b3&pli=1&cloudshell=true

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```
fortune-app-blue-5cd9c5fd96      3          3          3          23s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
fortune-app-blue-5cd9c5fd96-2pr76   1/1     Running   0           32s
fortune-app-blue-5cd9c5fd96-ng57r   1/1     Running   0           32s
fortune-app-blue-5cd9c5fd96-x9dws   1/1     Running   0           32s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl create -f services/fortune-app.yaml
service/fortune-app created
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27    <pending>        80:31269/TCP     10s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27    <pending>        80:31269/TCP     14s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27    <pending>        80:31269/TCP     16s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
fortune-app   LoadBalancer 34.118.233.27    34.11.254.223    80:31269/TCP     39s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ curl http://<EXTERNAL-IP>/^CrSION
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ curl http://34.11.254.223/version
{"version":"1.0.0"}
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ curl http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version
{"version":"1.0.0"}
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl scale deployment fortune-app-blue --replicas=5
deployment.apps/fortune-app-blue scaled
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods | grep fortune-app-blue | wc -l
5
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl scale deployment fortune-app-blue --replicas=3
deployment.apps/fortune-app-blue scaled
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods | grep fortune-app-blue | wc -l
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student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
```

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End Lab00:41:24

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Open Google Cloud Console

Username

student-02-e14d37b92bd3i

Password

Rf6aKqhuIWHB

Project ID

qwiklabs-gcp-04-f2da343i

Task 3. Rolling update

Deployments support updating images to a new version through a rolling update mechanism.

Trigger a rolling update

1. To trigger a rolling update, you can simply **apply** the configuration of the "green" deployment. Kubernetes is smart enough to see the existing deployment (fortune-app-blue) and will "roll" the changes from the new file onto it.

kubectl edit deployment fortune-app-blue

2. In the editor, find the **image** line and change the version tag from 1.0.0 to 2.0.0. You can edit the file by pressing **i** on your keyboard to enter "insert mode".

First, change the image tag:

Find this line: image: "us-central1-docker.pkg.dev/qwiklabs-

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student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl create -f services/fortune-app.yaml

service/fortune-app created

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get services fortune-app

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
fortune-app	LoadBalancer	34.118.233.27	<pending>	80:31269/TCP	10s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get services fortune-app

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
fortune-app	LoadBalancer	34.118.233.27	<pending>	80:31269/TCP	14s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get services fortune-app

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
fortune-app	LoadBalancer	34.118.233.27	<pending>	80:31269/TCP	16s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get services fortune-app

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
fortune-app	LoadBalancer	34.118.233.27	34.11.254.223	80:31269/TCP	39s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://<EXTERNAL-IP>/^CrSION

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://34.11.254.223/version

{ "version": "1.0.0" }

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version

{ "version": "1.0.0" }

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl scale deployment fortune-app-blue --replicas=5

deployment.apps/fortune-app-blue scaled

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get pods | grep fortune-app-blue | wc -l

5

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl scale deployment fortune-app-blue --replicas=3

deployment.apps/fortune-app-blue scaled

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get pods | grep fortune-app-blue | wc -l

3

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl edit deployment fortune-app-blue

error: deployments.apps "fortune-app-blue" is invalid

A copy of your changes has been stored to "/tmp/kubectl-edit-3132768138.yaml"

error: Edit cancelled, no valid changes were saved.

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl edit deployment fortune-app-blue

deployment.apps/fortune-app-blue edited

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$

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- Change value: "1.0.0" to value: "2.0.0"

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End Lab

00:36:30

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[Learn more.](#)

[Open Google Cloud Console](#)

Username

student-02-e14d37b92bd36

Password

Rf6aKqhuIWHB

Project ID

```
qwiklabs-gcp-04-f2da3430
```

3. Save and close the editor. You can do this by pressing `Esc`, then typing `:wq`, and pressing `Enter`. This will trigger the rolling update on the correct deployment and properly record its history. This will trigger the rolling update on the correct deployment and properly record its history.

4. See the new `ReplicaSet` that Kubernetes creates:

```
kubectl get replicaset
```

5. You can also see a new entry in the rollout history:

```
kubectl rollout history deployment/fortune-app-blue
```

Pause a rolling update

1. Run the following to pause the rollout:

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```
fortune-app LoadBalancer 34.118.233.27 <pending> 80:31269/TCP 16s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get services fortune-app
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
fortune-app LoadBalancer 34.118.233.27 34.11.254.223 80:31269/TCP 39s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ curl http://<EXTERNAL-IP>/^CrSION
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ curl http://34.11.254.223/version
{"version":"1.0.0"}
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ curl http://`kubectl get svc fortune-app -o=jsonpath="{.status.loadBalancer.ingress[0].ip}"`/version
{"version":"1.0.0"}
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl scale deployment fortune-app-blue --replicas=5
deployment.apps/fortune-app-blue scaled
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods | grep fortune-app-blue | wc -l
5
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl scale deployment fortune-app-blue --replicas=3
deployment.apps/fortune-app-blue scaled
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods | grep fortune-app-blue | wc -l
3
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl edit deployment fortune-app-blue
error: deployments.apps "fortune-app-blue" is invalid
A copy of your changes has been stored to "/tmp/kubectl-edit-3132768138.yaml"
error: Edit cancelled, no valid changes were saved.
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl edit deployment fortune-app-blue
deployment.apps/fortune-app-blue edited
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get replicaset
NAME DESIRED CURRENT READY AGE
fortune-app-blue-5cd9c5fd96 2 2 2 10m
fortune-app-blue-6d44ffb9cd 2 2 1 30s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl rollout history deployment/fortune-app-blue
deployment.apps/fortune-app-blue
REVISION CHANGE-CAUSE
1 <none>
2 <none>

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
```

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student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://<EXTERNAL-IP>/^CrSION

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://34.11.254.223/version

{ "version": "1.0.0" }

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version

{ "version": "1.0.0" }

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl scale deployment fortune-app-blue --replicas=5

deployment.apps/fortune-app-blue scaled

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get pods | grep fortune-app-blue | wc -l

5

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl scale deployment fortune-app-blue --replicas=3

deployment.apps/fortune-app-blue scaled

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get pods | grep fortune-app-blue | wc -l

3

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl edit deployment fortune-app-blue

error: deployments.apps "fortune-app-blue" is invalid

A copy of your changes has been stored to "/tmp/kubectl-edit-3132768138.yaml"

error: Edit cancelled, no valid changes were saved.

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl edit deployment fortune-app-blue

deployment.apps/fortune-app-blue edited

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get replicaset

NAME	DESIRED	CURRENT	READY	AGE
fortune-app-blue-5cd9c5fd96	2	2	2	10m
fortune-app-blue-6d44ffb9cd	2	2	1	30s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl rollout history deployment/fortune-app-blue

deployment.apps/fortune-app-blue

REVISION	CHANGE-CAUSE
1	<none>
2	<none>

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl rollout pause deployment/fortune-app-blue

deployment.apps/fortune-app-blue paused

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl rollout status deployment/fortune-app-blue

deployment "fortune-app-blue" successfully rolled out

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$

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End Lab00:35:11

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Open Google Cloud Console

Username

student-02-e14d37b92bd3i

Password

Rf6aKqhuIWHB

Project ID

qwiklabs-gcp-04-f2da343i

Note: The status command might immediately report "deployment "fortune-app-blue" successfully rolled out". This is expected and indicates that the pause command itself was successful. It does not mean the version update is complete.

3. Check the version of each pod. You'll see a mix of 1.0.0 and 2.0.0 pods, confirming the rollout is paused mid-way.

for p in \$(kubectl get pods -l app=fortune-app -o=jsonpath='{.items[*].metadata.name}'); do echo \$p && curl -s http://\$(kubectl get pod \$p -o=jsonpath='{.status.podIP}')/version; echo; done

Copied!

4. Press `Ctrl+C` to exit the loop.

Resume a rolling update

1. Continue the rollout using the `resume` command:

kubectl rollout resume deployment/fortune-app-blue

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```
{ "version": "1.0.0" }
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl scale deployment fortune-app-blue --replicas=5
deployment.apps/fortune-app-blue scaled
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods | grep fortune-app-blue | wc -l
5
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl scale deployment fortune-app-blue --replicas=3
deployment.apps/fortune-app-blue scaled
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods | grep fortune-app-blue | wc -l
3
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl edit deployment fortune-app-blue
error: deployments.apps "fortune-app-blue" is invalid
A copy of your changes has been stored to "/tmp/kubectl-edit-3132768138.yaml"
error: Edit cancelled, no valid changes were saved.
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl edit deployment fortune-app-blue
deployment.apps/fortune-app-blue edited
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get replicaset
NAME                                DESIRED  CURRENT  READY  AGE
fortune-app-blue-5cd9c5fd96          2         2         2     10m
fortune-app-blue-6d44ffb9cd          2         2         1     30s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl rollout history deployment/fortune-app-blue
deployment.apps/fortune-app-blue
REVISION  CHANGE-CAUSE
1          <none>
2          <none>

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl rollout pause deployment/fortune-app-blue
deployment.apps/fortune-app-blue paused
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl rollout status deployment/fortune-app-blue
deployment "fortune-app-blue" successfully rolled out
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ for p in $(kubectl get pods -l app=fortune-app -o=jsonpath='{.items[*].metadata.name}'); do echo $p && curl -s
http://$(kubectl get pod $p -o=jsonpath='{.status.podIP}')/version; echo; done
fortune-app-blue-6d44ffb9cd-2tbvz
^C
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
```

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End Lab

00:33:53

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Open Google Cloud Console

Username

student-02-e14d37b92bd3f

Password

Rf6aKqhuIWHB

Project ID

qwiklabs-gcp-04-f2da343f

Resume a rolling update

1. Continue the rollout using the resume command:

kubectl rollout resume deployment/fortune-app-blue

2. When the rollout is complete, you should see the following when running the status command:

kubectl rollout status deployment/fortune-app-blue

Roll back an update

Assume that a bug was detected in your new version.

1. Use the rollout command to roll back to the previous version:

30/100

deployments

Setup and requirements

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Task 1. Learn about the deployment object

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```
deployment.apps/fortune-app-blue scaled
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods | grep fortune-app-blue | wc -l
5
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl scale deployment fortune-app-blue --replicas=3
deployment.apps/fortune-app-blue scaled
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get pods | grep fortune-app-blue | wc -l
3
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl edit deployment fortune-app-blue
error: deployments.apps "fortune-app-blue" is invalid
A copy of your changes has been stored to "/tmp/kubectl-edit-3132768138.yaml"
error: Edit cancelled, no valid changes were saved.
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl edit deployment fortune-app-blue
deployment.apps/fortune-app-blue edited
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl get replicaset
NAME                                DESIRED    CURRENT    READY    AGE
fortune-app-blue-5cd9c5fd96         2          2          2        10m
fortune-app-blue-6d44ffb9cd         2          2          1        30s
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl rollout history deployment/fortune-app-blue
deployment.apps/fortune-app-blue
REVISION  CHANGE-CAUSE
1          <none>
2          <none>

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl rollout pause deployment/fortune-app-blue
deployment.apps/fortune-app-blue paused
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl rollout status deployment/fortune-app-blue
deployment "fortune-app-blue" successfully rolled out
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ for p in $(kubectl get pods -l app=fortune-app -o=jsonpath='{.items[*].metadata.name}'); do echo $p && curl -s
http://$(kubectl get pod $p -o=jsonpath='{.status.podIP}')/version; echo; done
fortune-app-blue-6d44ffb9cd-2tbvz
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student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl rollout resume deployment/fortune-app-blue
deployment.apps/fortune-app-blue resumed
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ kubectl rollout status deployment/fortune-app-blue
deployment "fortune-app-blue" successfully rolled out
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
```

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deployment.apps/fortune-app-blue edited

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl get replicaset

NAME	DESIRED	CURRENT	READY	AGE
fortune-app-blue-5cd9c5fd96	2	2	2	10m
fortune-app-blue-6d44ffb9cd	2	2	1	30s

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl rollout history deployment/fortune-app-blue

deployment.apps/fortune-app-blue

REVISION	CHANGE-CAUSE
1	<none>
2	<none>

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl rollout pause deployment/fortune-app-blue

deployment.apps/fortune-app-blue paused

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl rollout status deployment/fortune-app-blue

deployment "fortune-app-blue" successfully rolled out

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ for p in \$(kubectl get pods -l app=fortune-app -o=jsonpath='{.items[*].metadata.name}'); do echo \$p && curl -s http://\$(kubectl get pod \$p -o=jsonpath='{.status.podIP}')/version; echo; done

fortune-app-blue-6d44ffb9cd-2tbvz

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student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl rollout resume deployment/fortune-app-blue

deployment.apps/fortune-app-blue resumed

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl rollout status deployment/fortune-app-blue

deployment "fortune-app-blue" successfully rolled out

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl rollout undo deployment/fortune-app-blue

deployment.apps/fortune-app-blue rolled back

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version

{ "version": "2.0.0" }

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version

{ "version": "2.0.0" }

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ kubectl rollout undo deployment/fortune-app-blue

deployment.apps/fortune-app-blue rolled back

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version

{ "version": "1.0.0" }

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$ curl http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version

{ "version": "1.0.0" }

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) \$

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```
{ "version": "1.0.0" }
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $ cat deployments/fortune-app-canary.yaml
# orchestrate-with-kubernetes/kubernetes/deployments/fortune-app-canary.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: fortune-app-canary
spec:
  replicas: 1
  selector:
    matchLabels:
      app: fortune-app
  template:
    metadata:
      labels:
        app: fortune-app
        track: canary # The track is 'canary'
        version: "2.0.0"
    spec:
      containers:
        - name: fortune-app
          # The new, centralized image path for v2.0.0
          image: "us-central1-docker.pkg.dev/qwiklabs-resources/spl-lab-apps/fortune-service:2.0.0"
          ports:
            - name: http
              containerPort: 8080
          env:
            - name: APP_VERSION
              value: "2.0.0"
          resources:
            limits:
              cpu: "0.2"
              memory: "20Mi"
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3) $
```

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```
deployment.apps/fortune-app-canary created
student 02 e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
fortune-app-blue	3/3	3	3	15m
fortune-app-canary	1/1	1	1	8s

```
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$
```

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Managing Deployments Using Kubernetes Engine

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Managing Deployments Using Kubernetes Engine

Lab setup instructions and requirements

Protect your account and progress. Always use a private browser window and lab

End Lab00:29:49

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud Console

Username

student-02-e14d37b92bd3i

Password

Rf6aKqhuIWHB

Project ID

qwiklabs-gcp-04-f2da343i

1. You can verify the version being served by making requests to the service.

```
for i in {1..10}; do curl -s http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version; echo; done
```

2. Run this several times and you should see that most of the requests are served by version 1.0.0 and a small subset are served by 2.0.0.

Test completed task

Click **Check my progress** below to check your lab progress. If you successfully created Canary deployment, you'll see an assessment score.

Canary Deployment

Check my progress

Assessment Completed!

deployments

Setup and requirements

Set the zone

Get sample code for this lab

Task 1. Learn about the deployment object

Task 2. Create a deployment

Task 3. Rolling update

Task 4. Canary deployments

Task 5. Blue-green deployments

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```
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ for i in {1..10}; do curl -s http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version; echo; done
{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ for i in {1..10}; do curl -s http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version; echo; done
{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}

{"version":"2.0.0"}
```

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```
{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ for i in {1..10}; do curl -s http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version; echo; done
{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl apply -f services/fortune-app-blue-service.yaml
Warning: resource services/fortune-app is missing the kubect1.kubernetes.io/last-applied-configuration annotation which is required by kubect1 apply. kubect1 apply should only be used on res
ources created declaratively by either kubect1 create --save-config or kubect1 apply. The missing annotation will be patched automatically.
service/fortune-app configured
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl apply -f services/fortune-app-blue-service.yaml
service/fortune-app unchanged
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$
```

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```
kubectl apply -f services/fortune-app-blue-service.yaml
```

Updating using Blue-Green deployment

1. Now, create the new "green" deployment for version 2.0.0.

```
kubectl create -f deployments/fortune-app-green.yaml
```

2. Once the green deployment has started, verify that the current version being served is still 1.0.0.

```
curl http://`kubectl get svc fortune-app -o=jsonpath="{.status.loadBalancer.ingress[0].ip}"`/version
```

3. Now, update the service to point to the new "green" version:

```
kubectl apply -f services/fortune-app-green-service.yaml
```

Dashboard – qwiklabs-gcp-04-f2

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https://console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-f2da343063b3&pli=1&cloudshell=true

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Gemini CLI is available in Cloud Shell terminal! Type `gemini` to try it. [Learn more](#)

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```
{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

{ "version": "2.0.0" }

student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl apply -f services/fortune-app-blue-service.yaml
Warning: resource services/fortune-app is missing the kubectl.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on res
ources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
service/fortune-app configured
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl apply -f services/fortune-app-blue-service.yaml
service/fortune-app unchanged
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ kubectl create -f deployments/fortune-app-green.yaml
deployment.apps/fortune-app-green created
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$ curl http://`kubectl get svc fortune-app -o=jsonpath='{.status.loadBalancer.ingress[0].ip}'`/version
{ "version": "2.0.0" }
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{ "version": "2.0.0" }
student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$
```

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https://partner.skills.google/course_templates/625/labs/592705

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Google SkillsPartner

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53360

Develop Your Google Cloud Network

Managing Deployments Using Kubernetes Engine

Contents

Managing Deployments Using Kubernetes Engine

Lab setup instructions and requirements

Protect your account and progress. Always use a private browser window and lab

End Lab00:27:08

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud Console

Username

student-02-e14d37b92bd3f

Password

Rf6aKqhuIWHB

Project ID

qwiklabs-gcp-04-f2da343f

2. Once the green deployment has started, verify that the current version being served is still 1.0.0.

curl http://`kubectl get svc fortune-app -o=jsonpath="{.status.loadBalancer.ingress[0].ip}"`/version

3. Now, update the service to point to the new "green" version:

kubectl apply -f services/fortune-app-green-service.yaml

4. When the service is updated, the "green" deployment will be used immediately. You can now verify that version 2.0.0 is always being served:

curl http://`kubectl get svc fortune-app -o=jsonpath="{.status.loadBalancer.ingress[0].ip}"`/version

Blue-Green rollback

deployments

Setup and requirements

Set the zone

Get sample code for this lab

Task 1. Learn about the deployment object

Task 2. Create a deployment

Task 3. Rolling update

Task 4. Canary deployments

Task 5. Blue-green deployments

Congratulations!

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student_02_e14d37b92bd3@cloudshell:~/kubernetes (qwiklabs-gcp-04-f2da343063b3)$
```

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